

AI Techniques

- Search
- Use of Knowledge (Knowledge Representation and Reasoning)
- Abstraction

• Search:

Definition:

Search is a technique used in AI to find a solution by exploring different possible states or paths.

- Search algorithms can be broadly categorized as:
- **Uninformed Search (Blind Search):** These algorithms explore the search space without using any domain-specific knowledge or heuristics, such as Breadth-First Search (BFS) and Depth-First Search (DFS).
- **Informed Search (Heuristic Search):** These algorithms use heuristic functions to guide the search towards more promising states, such as A* Search and Greedy Best-First Search.
- **Adversarial Search:** Used in game-playing scenarios where an agent must consider the actions of an opponent, such as Minimax and Alpha-Beta Pruning.

• Use of Knowledge (Knowledge Representation and Reasoning):

Definition:

This technique involves storing knowledge in a structured form and using it to make decisions or solve problems.

- Different methods are used for knowledge representation:
- **Logical Representation:** Uses formal logic (e.g., propositional logic, first-order logic) to express knowledge and derive new conclusions through inference rules.
- **Semantic Networks:** Represent knowledge as a graph where nodes represent concepts and edges represent relationships between them.
- **Frame Representation:** Organizes knowledge into "frames" that represent typical objects or situations, with slots for attributes and relationships.
- **Production Rules (Rule-Based Systems):** Represent knowledge as IF-THEN rules, where actions are taken if certain conditions are met.

3. Abstraction

Definition:

Abstraction is the process of **reducing complexity by focusing only on important details and ignoring unnecessary information.**

Key Points:

- Simplifies complex problems
- Helps in efficient problem solving
- Used to create **models of real-world situations**
- Different levels of abstraction can be used

Advantages of Artificial Intelligence

- **High Accuracy with less errors:** AI machines or systems are prone to less errors and high accuracy as it takes decisions as per pre-experience or information.
- **High-Speed:** AI systems can be of very high-speed and fast-decision making, because of that AI systems can beat a chess champion in the Chess game.
- **High reliability:** AI machines are highly reliable and can perform the same action multiple times with high accuracy.
- **Useful for risky areas:** AI machines can be helpful in situations such as defusing a bomb, exploring the ocean floor, where to employ a human can be risky.
- **Digital Assistant:** AI can be very useful to provide digital assistant to the users such as AI technology is currently used by various E-commerce websites to show the products as per customer requirement.
- **Useful as a public utility:** AI can be very useful for public utilities such as a self-driving car which can make our journey safer and hassle-free, facial recognition for security purpose, Natural language processing to communicate with the human in human-language, etc.

Disadvantages of Artificial Intelligence

- **High Cost:** The hardware and software requirement of AI is very costly as it requires lots of maintenance to meet current world requirements.
- **Can't think out of the box:** Even we are making smarter machines with AI, but still they cannot work out of the box, as the robot will only do that work for which they are trained, or programmed.
- **No feelings and emotions:** AI machines can be an outstanding performer, but still it does not have the feeling so it cannot make any kind of emotional attachment with human, and may sometime be harmful for users if the proper care is not taken.
- **Increase dependency on machines:** With the increment of technology, people are getting more dependent on devices and hence they are losing their mental capabilities.
- **No Original Creativity:** As humans are so creative and can imagine some new ideas but still AI machines cannot beat this power of human intelligence and cannot be creative and imaginative.

Application of AI

1. Healthcare

AI is used to **diagnose diseases and assist doctors.**

- Medical diagnosis systems
- Predicting patient conditions
- Drug discovery

Example: Detecting cancer using medical images

2. Education

AI helps in **smart learning and personalized education.**

- Intelligent tutoring systems
- Automated grading
- Virtual classrooms

Example: Adaptive learning apps

3. Robotics

AI is used to control **intelligent robots.**

- Industrial robots
- Service robots
- Military robots

Example: Robots used in manufacturing



4. Natural Language Processing (NLP)

AI helps machines **understand and process human language**.

- Chatbots
- Language translation
- Speech recognition

Example: Voice assistants

5. Gaming

AI is used to create **smart and challenging game players**.

- Computer chess
- Strategy games

Example: AI beating human players in chess

6. Finance

AI is used in **banking and financial systems**.

- Fraud detection
- Stock market prediction
- Automated trading

7. Transportation

AI improves **transport systems and safety**.

- Self-driving cars
 - Traffic management
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8. Expert Systems

AI systems that **mimic human experts**.

- Used in medical, legal, and technical fields

